

The Euler line and special points in triangle geometry

Andrei Bud

1 Introduction

The motivation for this lecture comes from different directions:

1. First of all, we ask ourselves a simple question to better understand the configuration: *What methods appear in problems where we want to show that a point lies on the line OI , or the Euler line or the Euler circle?*

The answer is different compared to other situations: you are much more likely to hear an “I don’t know” or “it depends on the problem”—an answer that doesn’t appear as often when you ask experienced problem solvers about other configurations:

- Configurations with many circles passing through a single point? **Inversion...**
- Configurations with one circle tangent to another? **Archimedes’ Lemma, Protasov’s Lemma, Sawayama’s Lemma**, or properties of the **mixtilinear circle**...
- Configurations where at most one circle appears and all points are constructed via tangents or collinearities? **Projective geometry theorems**...
- Configurations involving completing a cyclic quadrilateral (possibly leading to a concurrency)? **Desargues’ Involution Theorem** and **cross ratios**...
- Collinearities and many points on a circle? **Pascal’s Theorem**...

The fact that we don’t have an intuition for problems involving the Euler line and Euler circle (especially the Euler line) is an atypical situation, and the goal of this course is to analyze problems of this type.

2. Another motivation is practical usefulness: problems of this type frequently appear in Olympiads, and in recent years also on IMO Shortlists (see IMO SHL 2023 G7 or IMO SHL 2022 G7).
3. Finally, the last motivation is the level of difficulty: these problems are more often than not, hard. As such, solving such a problem can be the deciding factor between getting a silver or gold medal.

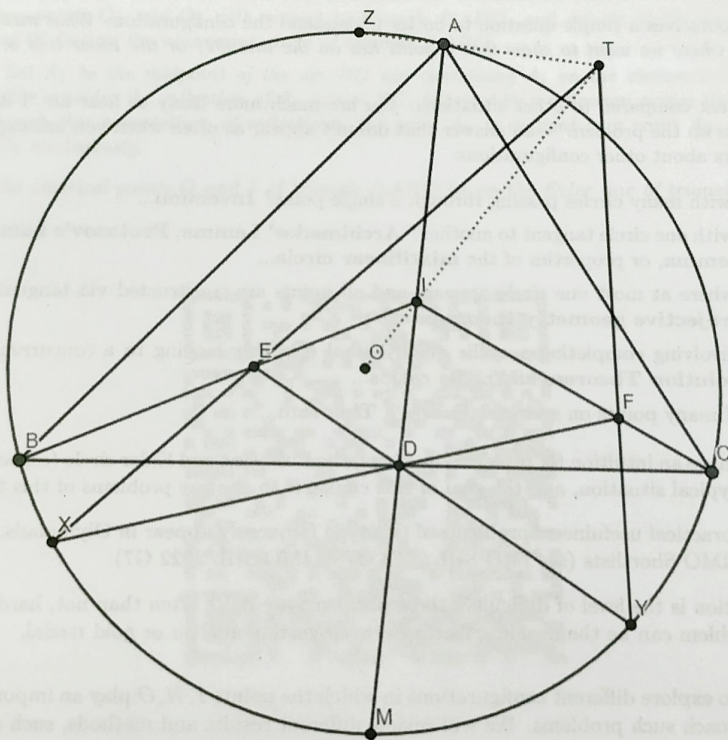
Our goal in this lecture is to explore different configurations in which the points I, H, O play an important role, and see how we can approach such problems. We will review different results and methods, such as

1. Desargues’ Involution Theorem and its dual;

2. Humpty-Dumpty points;
3. projecting the problem to a line using DIT;
4. relating harmonic divisions and power of a point.

2 Problems we will discuss

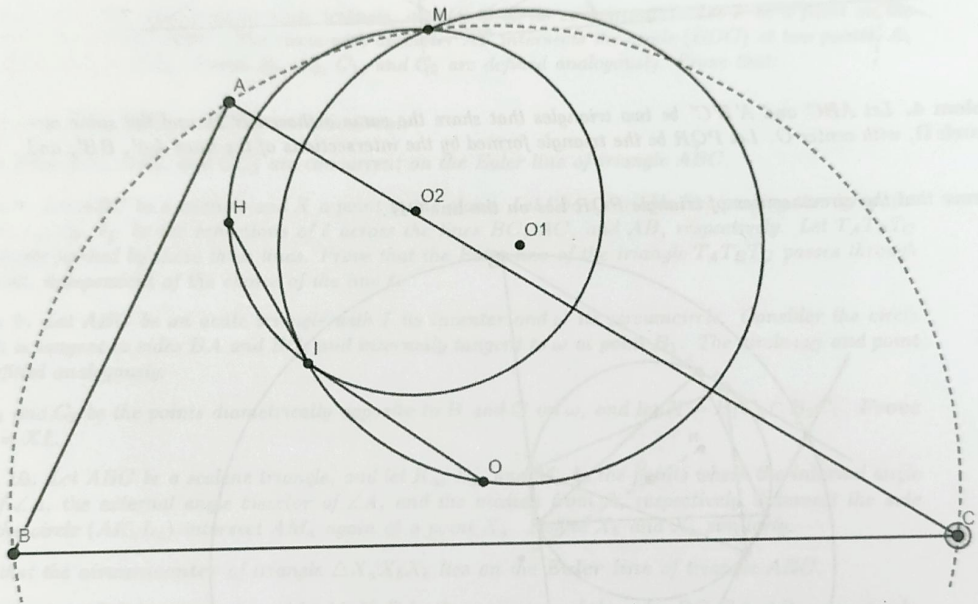
Problem 1. Let ABC be a triangle and let I and O be the incenter and circumcenter, respectively. Let X and Y be the midpoints of arcs ABC and ACB , respectively. The line AI intersects BC at D . Let E be the intersection of BI and DY , and similarly, let F be the intersection of CI and DX . Prove that the lines IO , EX , FY , and the external angle bisector of $\angle BAC$ are concurrent.



Problem 2. Let ABC be an arbitrary triangle, and let O , I , and H be the circumcenter, incenter, and orthocenter, respectively. Consider M as the second point of intersection between:

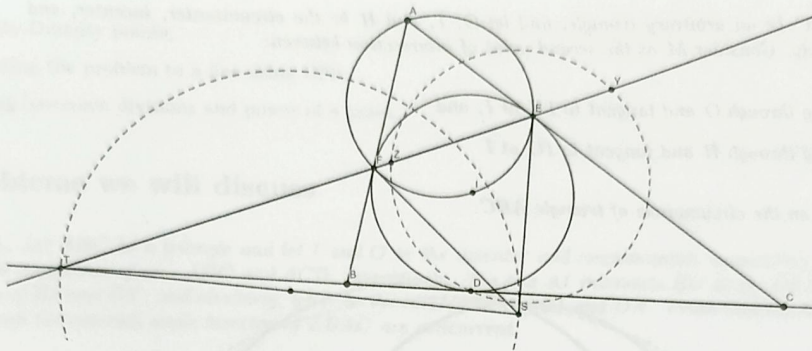
1. the circle passing through O and tangent to IH at I , and
2. the circle passing through H and tangent to IO at I .

Prove that M lies on the circumcircle of triangle ABC .



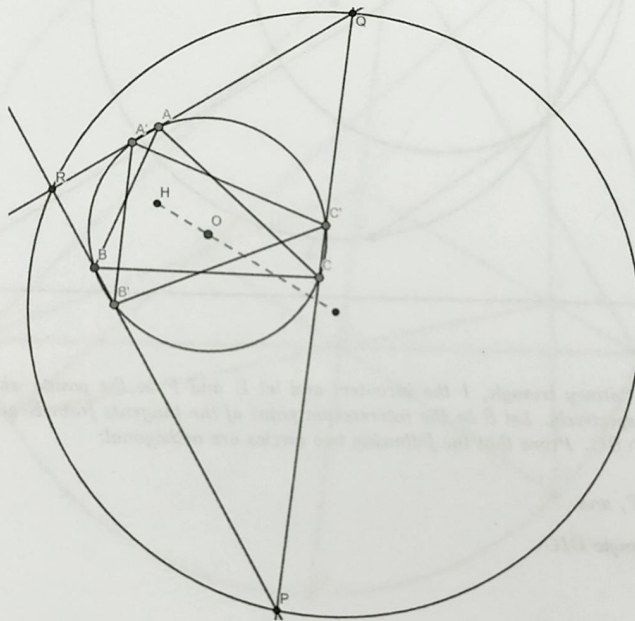
Problem 3. Let ABC be an arbitrary triangle, I the incenter, and let E and F be the points where the incircle touches AC and AB , respectively. Let S be the intersection point of the tangents from E and F to the circle (AEF) . Let $T = EF \cap BC$. Prove that the following two circles are orthogonal:

1. the circle with diameter ST , and
2. the nine-point circle of triangle BIC .



Problem 4. Let ABC and $A'B'C'$ be two triangles that share the same orthocenter H and the same circumcircle Ω , with center O . Let PQR be the triangle formed by the intersections of the lines AA' , BB' , and CC' .

Prove that the circumcenter of triangle PQR lies on the line OH .



3 Handout

Problem 5. Let ABC be a scalene triangle with H its orthocenter, and let Ω be its circumcircle, centered at O . Let M and N be the midpoints of sides AB and AC , respectively, and let E and F be the feet of the altitudes from B and C in triangle ABC . Let P be the intersection of MN with the tangent to Ω at A . Let Q be the second point of intersection of Ω and the circle (AEF) . Let $R = AQ \cap EF$. Prove that $PR \perp OH$.

Problem 6. Let ABC be a scalene triangle and O be its circumcenter. Let P be an arbitrary point on circle (BOC) . Let D and E be the projections of P onto the sides AB and AC , respectively. Prove that the orthocenter of $\triangle ADE$ lies on the nine-point circle of $\triangle ABC$.

Problem 7. Let ABC be a scalene acute triangle, and let O be its circumcenter. Let T be a point on the line BC such that $\angle TAO = 90^\circ$. The circle with diameter $\overline{AT}$ intersects the circle (BOC) at two points, A_1 and A_2 , where $OA_1 < OA_2$. Points B_1, B_2, C_1 , and C_2 are defined analogously. Prove that:

- The lines $\overline{AA_1}, \overline{BB_1}$, and $\overline{CC_1}$ are concurrent.
- The lines $\overline{AA_2}, \overline{BB_2}$, and $\overline{CC_2}$ are concurrent on the Euler line of triangle ABC .

Problem 8. Let ABC be a triangle and X a point in the plane. Let ℓ be a variable line passing through point X , and let ℓ_A, ℓ_B, ℓ_C be the reflections of ℓ across the lines BC, AC , and AB , respectively. Let $T_A T_B T_C$ be the triangle formed by these three lines. Prove that the Euler line of the triangle $T_A T_B T_C$ passes through a fixed point, independent of the choice of the line ℓ .

Problem 9. Let ABC be an acute triangle with I its incenter and ω its circumcircle. Consider the circle ω_B , which is tangent to sides BA and BC , and internally tangent to ω at point B_1 . The circle ω_C and point C_1 are defined analogously.

Let B_2 and C_2 be the points diametrically opposite to B and C on ω , and let $X = B_1 C_2 \cap B_2 C_1$. **Prove that $XA = XI$.**

Problem 10. Let ABC be a scalene triangle, and let K_a, L_a , and M_a be the points where the internal angle bisector of $\angle A$, the external angle bisector of $\angle A$, and the median from A , respectively, intersect the side BC . Let the circle $(AK_a L_a)$ intersect AM_a again at a point X_a . Define X_b and X_c similarly.

Prove that the **circumcenter** of triangle $\triangle X_a X_b X_c$ lies on the **Euler line** of triangle ABC .

Problem 11. Let ABC be a triangle and let M, N, P be the midpoints of the sides BC, CA, AB , respectively. Consider Ω_a , the circle passing through N and P , tangent to the circumcircle of triangle ABC at a point T_a lying on the arc BC that does **not** contain A . We define the circles Ω_b, Ω_c and the points T_b, T_c analogously.

Prove that the lines AT_a, BT_b , and CT_c are concurrent on the **Euler line** of triangle $\triangle ABC$.

Problem 12. Let ABC be a triangle, and let d be its Euler line. Consider two points E and F on AC and AB , respectively, such that $EF \perp d$. Let E' be the reflection of E across AB , and let F' be the reflection of F across AC . Define the points:

$$P = EE' \cap FF', \quad Q = E'F \cap EF', \quad \text{and} \quad R = PQ \cap EF.$$

Prove that $AR \perp BC$.

Problem 13. Let ABC be a triangle with I and O denoting its incenter and circumcenter, respectively. Let ω_A be the circle passing through B and C that is tangent to the incircle of triangle ABC . Define ω_B and ω_C analogously. Let A' be the second point of intersection of ω_B and ω_C , and define B' and C' similarly. Prove that the lines AA' , BB' , and CC' concur at a point lying on the line OI .

Problem 14. Let $\triangle ABC$ be a triangle and let D_A be the associated A -Dumpty point. Let D, E, F be the reflections of D_A across BC, CA , and AB , respectively. Let the nine-point circles of triangles $\triangle ABC$ and $\triangle DEF$ intersect at points X and Y . Prove that the points A, D_A, X, Y are concyclic.

Problem 15. Let ABC be a scalene triangle with $BC > AC > AB$. Let $P_1 \neq P_2$ be two points in the plane satisfying the following condition:

★ For $i = 1, 2$, if we denote by D_i, E_i, F_i the intersections of lines AP_i, BP_i, CP_i with the circumcircle (ABC) , then

$$D_i E_i \perp D_i F_i \quad \text{and} \quad D_i E_i = D_i F_i \neq 0.$$

Let Q_1, Q_2 be the points where the line $P_1 P_2$ intersects the circumcircle (ABC) . Consider the Simson lines of these two points Q_1 and Q_2 with respect to triangle $\triangle ABC$, and denote by W their intersection point. Prove that W lies on the nine-point circle of triangle $\triangle ABC$.

Problem 16. Let A_1 be the midpoint of the arc BC not containing A , on the circumcircle of the acute triangle ABC . We consider the reflection of A_1 across BC , followed by a reflection across the angle bisector of $\angle BAC$. Through this composition of reflections, the point A_1 is mapped to a point A_2 . We define the points B_2 and C_2 analogously.

Prove that the classical points O and I of triangle $\triangle ABC$ lie on the Euler line of triangle $A_2 B_2 C_2$.



Figure 1: Solutions to the problems